

Wookyeong Song

CONTACT INFORMATION

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EMPLOYMENT

Brown University, Providence, RI

Brown Data Science Institute and Department of Biostatistics
Postdoctoral Research Associate, 2026 -
Advisor: [Professor Larry Han](#)

EDUCATION

University of California, Davis, CA

Ph.D. in Statistics, 2021 - 2026
Advisor: [Professor Hans-Georg Müller](#)
Thesis: Curvature Inference and Interpretable Regression for Random Objects in Metric Spaces

Seoul National University, Seoul, South Korea

B.S. in Statistics and Mathematical Sciences (Double Major), 2021
Advisor: [Professor Hee-Seok Oh](#) & [Professor Taesung Park](#)
• Grade: Summa cum laude

RESEARCH INTEREST

Metric geometry and statistics; Manifold learning; Optimal transport; Uncertainty Quantification; Functional data analysis; Nonparametric statistics; Survival Analysis; Artificial intelligence; Brain networks

PUBLICATIONS & PREPRINTS

1. **Song, W.**, Dubey, P., Müller, H.-G., & Petersen, A. (2026). Inference for Fréchet regression, *Under Revision*. [[preprint](#)]
2. Choi, C.[†], **Song, W.**[†], Müller, H.-G., & Park, B. U. (2026). Additive Fréchet regression of random objects, *Under Revision*. [†]Equal Contribution
3. **Song, W.**[†], Zhou, H.[†], Zhou, Y.[†] & Müller, H.-G. (2026). Non-Euclidean data analysis with metric statistics, *Harvard Data Science Review, Accepted*. [[paper](#)] [[code](#)]
4. **Song, W.** & Müller, H.-G. (2026). ADOPT: Additive optimal transport regression, *AISTATS, Accepted*. [[paper](#)][[code](#)]
5. **Song, W.** & Müller, H.-G. (2026). Inference for dispersion and curvature of random objects, *Journal of the American Statistical Association*, 121(553), 729-740. [[paper](#)][[code](#)]
 - AISTATS 2026 Journal-to-Conference Track
 - Student Paper Award Finalist by ASA Nonparametric Statistics Section, JSM 2024
6. **Song, W.**, Oh, H.-S., Cheung, K. & Lim, Y. (2024). Multi-feature clustering of step data using multivariate functional principal component

analysis,
Statistical Papers, 65(4), 2109-2134. [\[paper\]](#)[\[code\]](#)

7. Kim, H.[†], **Song, W.**[†], Choo, W., Lee, S., ..., Park, T., & Jang, J.-Y. (2023)
 Development, validation, and comparison of a nomogram based on radiologic findings
 for predicting malignancy in intraductal papillary mucinous neoplasms of the pancreas:
 An international multicenter study,
Journal of Hepato-Biliary-Pancreatic Sciences, 30(1), 133-143. [\[paper\]](#)
8. Kang, J., Lee, C., **Song, W.**, Choo, W., ..., Park, T., & Jang, J.-Y. (2020)
 Risk prediction for malignant intraductal papillary mucinous neoplasm of the pancreas:
 logistic regression versus machine learning,
Scientific Reports, 10, 20140. [\[paper\]](#)

TALKS

Inference for Fréchet regression

- Joint Statistical Meeting (JSM), Boston, MA (Aug 2026)

ADOPT: Additive optimal transport regression

- The 29th International Conference on Artificial Intelligence and Statistics (AISTATS), Tangier, Morocco. (May 2026)

Inference for dispersion and curvature of random objects

- AISTATS, Journal-to-Conference Track, Tangier, Morocco. (May 2026)
- Joint Statistical Meetings (JSM), Topic-contributed Paper Session, Portland, OR. (Aug 2024)
- Statistics in the Age of AI, George Washington University, DC. (May 2024)
- Princeton Machine Learning Theory Summer School, Princeton University, NJ. (Jun 2023)

Multi-feature clustering of step data using multivariate functional principal component analysis

- Fall Korean Statistical Society Conference, University of Seoul, South Korea. (Nov 2019)

HONORS & AWARDS

UC Davis Graduate Studies Travel Award	2026
IMS Hannan Graduate Student Travel Award, <i>Institute of Mathematical Statistics</i>	2026
Peter G. Hall Award, <i>Department of Statistics, UC Davis</i>	2025
Best Presentation Award, <i>ASA Nonparametric Statistics Section</i>	2024
Student Paper Award Finalist, <i>ASA Nonparametric Statistics Section</i>	2024
Travel Award, <i>Princeton Machine Learning Theory Summer School</i>	2023
Julius Blum Award, <i>Department of Statistics, UC Davis</i>	2022
Excellent Tutoring Award, <i>Seoul National University</i>	2020
Dean's List, <i>College of Natural Science, Seoul National University</i>	2020
3 rd Prize, <i>Poster presentation, Fall Korean Statistical Society Conference</i>	2019

TEACHING EXPERIENCE

University of California, Davis Associate Instructor

- STA103: Applied Statistics for Business and Economics (Summer 2025)
 (Overall Rating: 4.75/5.0)

Teaching Assistant

	• STA010: Statistical Thinking (Fall 2021, Spring 2022) • STA142A: Statistical Learning I (Winter 2022) • STA141B: Data & Web Technologies for Data Analysis (Fall 2022) • STA206: Statistical Methods for Research I (Fall 2022) • STA100: Applied Statistics for Biological Science (Winter 2023) • STA013: Elementary Statistics (Spring 2023, Spring 2026) • STA103: Applied Statistics for Business and Economics (Spring 2025) • STA104: Nonparametric Statistics (Fall 2025) • STA138: Categorical Data Analysis (Winter 2026)	
MENTORING	Muqing Cui (now Ph.D. student at UC Davis) Project: Fréchet variance process	2023 - Present
SERVICE	Referee Journal of Machine Learning Research (1), Journal of the American Statistical Association (1), Biometrika (3), Harvard Data Science Review (2) Conference Organization • Chair, “Functional Data Analysis: Regression, Latent Structure, and Inference” session, Joint Statistical Meetings, Boston, MA. (Aug 2026)	
PROFESSIONAL EXPERIENCE	Capital One, Plano, TX Data Science Ph.D. Intern • Developed explainable AI (XAI) models to estimate risk of account-level auto loans, ensuring transparency and optimized performance. Republic of Korea Army, South Korea Sergeant	Jun 2024 - Aug 2024 Feb 2016 - Nov 2017
SOFTWARE	frechet Statistical Analysis for Random Objects and Non-Euclidean Data (R package on GitHub).	